

Graduate Algebra
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1. Week 1 [2026-09-09]

We'll begin by recalling some basic sorts of algebra that you more-or-less encountered before.

1.1. Notations and recollections

We reserve the following letters:

- $\mathbb{N}$ for the set of *natural numbers* $0, 1, 2, \dots$
- $\mathbb{Z}$ for the set of *integers*, i.e. for all $\pm n$ for $n \in \mathbb{N}$
- $\mathbb{Q}$ for the set of *rational numbers* m/n for $m, n \in \mathbb{Z}$ with $n \neq 0$
- $\mathbb{R}$ for the set of *real numbers*, and
- $\mathbb{C}$ for the set of *complex numbers* $a + bi$ for $a, b \in \mathbb{R}$.

In this first lecture, I want to recall some of the main objects of study in algebra, including: groups, rings and fields. Ultimately, the goal today is to prove an analogue of Cayley's Theorem - see Theorem 1.6.1 and Theorem 1.7.1 about embedding arbitrary groups in some standard groups.

1.2. Groups

Recall that a group is a set G together with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following:

- associativity: $\forall x, y, z \in G, (xy)z = x(yz)$
- identity: $\exists e \in G, xe = ex = x$.
- inverses: $\forall x \in G, \exists y \in G, xy = yx = 1$.

Remark 1.2.1:

- We usually write 1 or sometimes 1_G rather than e for the identity element of G .
- we usually write x^{-1} for the inverse of $x \in G$
- there are *uniqueness* results that I'm eliding here; the identity 1 of G is unique, and the inverse x^{-1} of an element is unique. These statements are *consequences* of the above axioms (they don't require additional assumption.)
- A group is abelian if $\forall a, b \in G, ab = ba$
- Sometimes we write groups additively; in that case, 0 is the identity element and the inverse of $a \in G$ is $-a \in G$. We always insist that additive groups are abelian.

Definition 1.2.2: For groups G and H , a function $\varphi : G \rightarrow H$ is a **group homomorphism** provided that $\forall x, y \in G, \varphi(xy) = \varphi(x)\varphi(y)$.

Definition 1.2.3: Let $\varphi : G \rightarrow H$ be a group homomorphism. The **kernel** of φ is

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1.\}$$

Remark 1.2.4: If $\varphi : G \rightarrow H$ is a group homomorphism, $\ker \varphi$ is a subgroup of G - i.e. $\ker \varphi$ is non-empty, and is closed under multiplication and under taking inverses.

Proposition 1.2.5: Let $\varphi : G \rightarrow H$ be a group homomorphism. Then φ is an injective (or one-to-one) function if and only if $\ker \varphi = \{1_G\}$.

1.3. Rings

Definition 1.3.1: A **ring** is an additive abelian group R together with a binary operation of multiplication

$$\cdot : R \times R \rightarrow R$$

which satisfies the following:

- multiplication is associative: $\forall a, b, c \in R, (ab)c = a(bc)$.
- there is a multiplicative identity: $\exists 1 \in R, \forall a \in R, 1a = a1 = a$.
- distribution laws: $\forall a, b, c \in R, a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

The ring R is **commutative** provided that $\forall a, b \in R, ab = ba$.

Example 1.3.2:

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative rings.
- For a natural number $n > 1$, the ring $\text{Mat}_n(\mathbb{Z})$ of $n \times n$ matrices with coefficients in $\mathbb{Z}$ is a non-commutative ring.

Definition 1.3.3: For a commutative ring R , an element $a \in R$ is a **unit** provided that $\exists v \in R, uv = vu = 1$.

The set $R^\times$ of units in R is a group under the multiplication of R .

1.4. Fields

Definition 1.4.1: A **field** is a commutative ring F such that $\forall a \in F, a \neq 0 \Rightarrow a$ is a unit.

Example 1.4.2: $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields, but $\mathbb{Z}$ is not a field.

Remark 1.4.3: A possibly non-commutative ring satisfying the requirement $\forall a \in F, a \neq 0 \Rightarrow a$ is a unit is known as a **division ring**. The most well-known example of a non-commutative division ring is Hamilton's quaternions $\mathbb{H}$: $\mathbb{H}$ is a 4 dimensional $\mathbb{R}$ -vector space with basis $1, i, j, k$ where the basis vectors multiply according to the rules $i^2 = j^2 = k^2 = -1, ij = -ji = k, jk = -kj = i$ and $ki = -ik = j$.

If $0 \neq v = a + bi + cj + dk \in \mathbb{H}$, then $v^{-1} = (a - bi - cj - dk)/(a^2 + b^2 + c^2 + d^2)$.

1.5. Linear Algebra

Definition 1.5.1: If F is a field, a vector space over F – or an F -vector space – is an additive abelian group V together with an operation of scalar multiplication

$$F \times V \rightarrow V$$

written $(t, v) \mapsto tv$, subject to the following:

- identity: $\forall v \in V, 1v = v$.
- associativity: $\forall a, b \in F$ and $v \in V, a(bv) = (ab)v$.
- distributive laws: $\forall a, b \in F$ and $v, w \in V, (a + b)v = av + bv$ and $a(v + w) = av + aw$.

Remark 1.5.2: Probably in a linear algebra class you saw results stated for vector spaces over $\mathbb{R}$ or $\mathbb{C}$; however, many general results in linear algebra remain valid for vector space over F .

Example 1.5.3: Let I be any set, and let V be the set of all functions $f : I \rightarrow F$ which have finite support. Recall that the support of f is $\{x \in I \mid f(x) \neq 0\}$.

Then V is a vector space. (The addition and scalar multiplication operations are define “pointwise” – see homework.)

Remark 1.5.4: Recall that a **basis** of a vector space is subset B of V which is linearly indepent and spans V . The vector space of finitely supported functions $I \rightarrow F$ has a basis $B = \{\delta_i \mid i \in I\}$, where

$$\delta_i : I \rightarrow F$$

is the function defined by $\delta_i(j) = 0$ if $i \neq j$ and $\delta_i(i) = 1$.

Definition 1.5.5: If V and W are F -vector spaces, an F -linear map $\varphi : V \rightarrow W$ is a homomorphism of additive groups which satisfies the condition

$$\forall t \in F, \forall v \in V, \varphi(tv) = t\varphi(v).$$

Definition 1.5.6: If V is an F -vector space, the general linear group $\text{GL}(V)$ is the set

$$\{\varphi : V \rightarrow V \mid \varphi \text{ is } F\text{-linear and invertible.}\}$$

$\text{GL}(V)$ is a group whose operation is given by composition of linear transformations.

Remark 1.5.7: If V is finite dimensional, so that V is isomorphic to F^n as F -vector spaces, linear algebra shows that $\text{GL}(V)$ is isomorphic to the group GL_n of $n \times n$ matrices with non-zero determinant, where $n = \dim_F V$ and where the operation in GL_n is given by matrix multiplication.

Example 1.5.8: If I is a finite set with $n \in \mathbb{N}$ elements, then the vector space of finitely supported F -valued functions on I is isomorphic to the space F^n of column vectors.

Remark 1.5.9: In fact, if D is a division ring one can consider vector spaces over D . Rather than make an ad hoc definition for this, we will later define the notion of a module over a ring. A vector space over a field F is just an F -module, and then a vector space over a division algebra D is just a D -module.

1.6. Cayley's Theorem

Let Ω be any set. The set $S(\Omega)$ of all bijective functions $\psi : \Omega \rightarrow \Omega$ is a group whose operation is composition of functions.

Theorem 1.6.1 (Cayley's Theorem): Let G be any group. Then G is isomorphic to a subgroup of $S(\Omega)$ for some Ω .

Proof: Let $\Omega = G$. For $g \in G$, define a mapping $\lambda_g : G \rightarrow G$ by the rule

$$\lambda_g(h) = gh.$$

We are going to argue that the mapping $g \mapsto \lambda_g$ defines an injective group homomorphism

$$G \rightarrow S(\Omega) = S(G).$$

First of all, we note that $\lambda_1 = \text{id}$. Indeed, to check this identity of functions, let $h \in \Omega = G$. Then

$$\lambda_1(h) = 1h = h = \text{id}(h);$$

this confirms $\lambda_1 = \text{id}$.

Next, we note that for $g_1, g_2 \in G$, we have $(*) \quad \lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2}$. Again, to confirm this identity of functions, we let $h \in \Omega = G$. Then

$$(\lambda_{g_1} \circ \lambda_{g_2})h = \lambda_{g_1}(\lambda_{g_2}(h)) = \lambda_{g_1}(g_2 h) = g_1(g_2 h) = (g_1 g_2)h = \lambda_{g_1 g_2}(h)$$

as required.

Now, using $(*)$ we see for $g \in G$ that $\lambda_g \circ \lambda_{g^{-1}} = \lambda_1 = \text{id} = \lambda_{g^{-1}} \circ \lambda_g$, which proves that λ_g is bijective; thus indeed $\lambda_g \in S(\Omega) = S(G)$.

Moreover, $(*)$ shows that the mapping $\lambda : G \rightarrow S(G)$ given by $g \mapsto \lambda_g$ is a group homomorphism.

It remains to see that λ is injective. If $g \in \ker \lambda$, then $\lambda_g = \text{id}$. Thus $1 = \text{id}(1) = \lambda_g(1) = g1 = g$. Thus $g = 1$ so that $\ker \lambda = \{1\}$ which confirms that λ is injective by Proposition 1.2.5. This completes the proof. ■

1.7. A linear analogue of Cayley's Theorem.

Let F be a field.

Theorem 1.7.1: Let G be any group. Then G is isomorphic to a subgroup of $\text{GL}(V)$ for some F -vector space V .

Proof: The proof is quite similar to the proof of Cayley's Theorem.

Let V be the vector space of all finitely supported functions $f : G \rightarrow F$. Recall that V has a basis $B = \{\delta_g \mid g \in G.\}$

We are going to define an injective group homomorphism $G \rightarrow \text{GL}(V)$.

For $g \in G$, we may define an F -linear mapping $\lambda_g : V \rightarrow V$ by specifying the value of λ_g at each vector in B . We set $\lambda_g(\delta_h) = \delta_{gh}$ for each $h \in G$.

Recall that a typical element v of V has the form

$$v = \sum_{i=1}^n t_i \delta_{h_i}$$

for scalars $t_i \in F$ and elements $h_i \in G$; since λ_g is F -linear, we have

$$\lambda_g(v) = \sum_{i=1}^n t_i \delta_{gh_i}.$$

We now show that $\lambda_1 = \text{id}$. To prove this, since the functions $V \rightarrow V$ are linear, it is enough to argue that the functions agree at each element of the basis B of V . Well, for $h \in G$,

$$\lambda_1(\delta_h) = \delta_{1h} = \delta_h = \text{id}(\delta_h)$$

as required.

We next show for $g_1, g_2 \in G$ that $(*)$ $\lambda_{g_1} \circ \lambda_{g_2} = \lambda_{g_1 g_2}$. Again, it suffices to argue that these functions agree at each element δ_h of B . For $h \in G$ we have:

$$(\lambda_{g_1} \circ \lambda_{g_2})(\delta_h) = \lambda_{g_1}(\lambda_{g_2} \delta_h) = \lambda_{g_1}(\delta_{g_2 h}) = \delta_{g_1(g_2 h)} = \delta_{(g_1 g_2)h} = \lambda_{g_1 g_2} \delta_h$$

as required.

Now, for $g \in G$ we see that by $(*)$ that

$$\text{id} = \lambda_1 = \lambda_g \circ \lambda_{g^{-1}}$$

which proves that λ_g is invertible and hence in $\text{GL}(V)$.

Moreover, (*) shows that the assignment $\lambda : G \rightarrow \text{GL}(V)$ given by the rule $g \mapsto \lambda_g$ is a group homomorphism.

It remains to argue that λ is injective. Suppose that $x \in \ker \lambda$, so that $\text{id} = \lambda_x$.

Then $\delta_1 = \text{id}(\delta_1) = \lambda_x(\delta_1) = \delta_{x1} = \delta_x$. This implies that $1 = x$ so that indeed the kernel of λ is trivial and thus λ is injective by Proposition 1.2.5.

■

2. Week 2 [2026-09-14]

2.1. Quotient of a set by an equivalence relation

Let S be a set and let $\sim$ be an equivalence relation on S – recall this means $\sim$ is **reflexive**, **symmetric**, and **transitive**: $\forall s, s \sim s$; $s_1 \sim s_2 \Rightarrow s_2 \sim s_1$; and $s_1 \sim s_2$ and $s_2 \sim s_3 \Rightarrow s_1 \sim s_3$.

Definition 2.1.1: For $s \in S$, the **equivalence class** $[s]$ is the subset

$$[s] = \{x \in S \mid x \sim s\}.$$

Proposition 2.1.2: For arbitrary $s, s' \in S$:

- a. $s \sim s' \Leftrightarrow [s] = [s']$
- b. $[s] \neq [s'] \Leftrightarrow [s] \cap [s'] = \emptyset$

Proof: Review! ■

Proposition 2.1.3: S is the disjoint union of its equivalence classes.

Proof: Each $x \in S$ lies in $[x]$. If $z \in [x] \cap [y]$ then $x \sim z$ and $y \sim z$, so $x \sim y$ by transitivity and $[x] = [y]$. ■

Definition 2.1.4: The **quotient** of S by $\sim$ is the pair $(\bar{S}, \pi)$ where $\bar{S} = \{[s] \mid s \in S\}$ and $\pi : S \rightarrow \bar{S}$ is given by $\pi(s) = [s]$.

By construction, π is surjective, and Proposition 2.1.2 (a) gives $a \sim b \Rightarrow \pi(a) = \pi(b)$ (with equality forced **only** in that case).

Remark 2.1.5: For $x \in S$, $\pi^{-1}(\pi(x)) = [x]$: this is immediate from Proposition 2.1.2 (a).

Proposition 2.1.6: (Universal property of the quotient) Let T be a set and $f : S \rightarrow T$ a function with $\forall a, b \in S, a \sim b \Rightarrow f(a) = f(b)$. Then there is a **unique** function $\bar{f} : \bar{S} \rightarrow T$ with $f = \bar{f} \circ \pi$.

Proof: Proposition 2.1.2 (a) shows the rule $[s] \mapsto f(s)$ is well-defined, giving $\bar{f}$ with $\bar{f} \circ \pi = f$. Since π is surjective, any function satisfying this equation already agrees with $\bar{f}$ everywhere, so $\bar{f}$ is unique. ■

Remark 2.1.7: Proposition 2.1.6 is really what characterizes $(\bar{S}, \pi)$: applying it in both directions to two quotients of the same $\sim$ produces mutually inverse maps between them, so any two quotients are canonically isomorphic. (Good exercise – write this out.) It's harmless, then, to work with **the** quotient $S/\sim$ via whichever concrete model is convenient – here, equivalence classes.

Readers who've seen Lean's Quot/Quotient type will recognize this shape: `Quot.mk` is π , `Quot.lift` builds $\bar{f}$ from f plus a proof that f respects $\sim$, and `Quot.sound` is Proposition 2.1.2 (a).

2.2. Sub-groups

Let G be a group (when giving definitions, we'll write G multiplicatively).

Definition 2.2.1: A **subgroup** of G is a non-empty subset $H \subseteq G$ such that H is closed under the operations of multiplication in G and inversion in G . In other words,

$$\forall a, b \in G, ab \in H \text{ and } a^{-1} \in H$$

.

Example 2.2.2: Consider the group $G = \mathbb{Z} \times \mathbb{Z}$ where the operation is componentwise addition. Check the following!

- $H_1 = \{(a, b) \in G \mid 2a + 3b = 0\}$ is a subgroup.
- $H_2 = \{n(2, 2) + m(1, 2) \mid n, m \in \mathbb{Z}\}$ is a subgroup.

The collection of subgroups of G has a natural partial order given by *containment*.

Proposition 2.2.3: (Constructing subgroups)

- If H_i for $i \in I$ is a family of subgroups of G , indexed by some set I , then the intersection $\bigcap_{i \in I} H_i$ is again a subgroup of G .
- Let $S \subseteq G$ be a subset. There is a unique smallest subgroup $H(S) = \langle S \rangle$ containing S . In other words, for any subgroup H' of G with $S \subseteq H'$, we have $\langle S \rangle \subseteq H'$.

Remark 2.2.4:

- If $S, T \subseteq G$ are subsets, we often write $\langle S, T \rangle$ for $\langle S \cup T \rangle$. If $S = \{s_1, s_2, \dots, s_n\}$ we often write $\langle S \rangle = \langle s_1, s_2, \dots, s_n \rangle$.
- The subgroup in Example 2.2.2(b) is precisely $\langle (2, 2), (1, 2) \rangle$.
- For any group G and $a \in G$, $\langle a \rangle := \langle \{a\} \rangle$ is the **cyclic subgroup** generated by a . If G is multiplicative, then $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ while if G is additive then $\langle a \rangle = \{na \mid n \in \mathbb{Z}\}$.

2.3. Group actions

Definition 2.3.1: Let G be a group and let Ω be a set. An **action** of G on Ω is a mapping

$$G \times \Omega \rightarrow \Omega \text{ written } (g, x) \mapsto gx$$

such that for each $x \in \Omega$ we have

- $1x = x$
- $\forall g, h \in G, (gh)x = g(hx)$.

For brevity, sometimes we say that Ω is a G -space.

Proposition 2.3.2: An action of a group G on a set Ω determines a homomorphism $f : G \rightarrow S(\Omega)$ such that $f(g)(x) = gx$ for $g \in G$ and $x \in \Omega$.

Conversely, given a homomorphism $f : G \rightarrow S(\Omega)$, there is an action of G on Ω given by $gx = f(g)(x)$ for each $g \in G$ and $x \in \Omega$.

Definition 2.3.3: Suppose that Ω is a G -space. The G -conjugacy relation on Ω is defined as follows: for $x, y \in \Omega$, $x \sim_G y$ provided that $\exists g \in G, gx = y$.

Proposition 2.3.4: The G -conjugacy relation on Ω is an equivalence relation.

Definition 2.3.5: Let Ω be a G -space, and let $\varphi : \Omega \rightarrow \Omega / \sim$ be the quotient mapping for the G -conjugacy relation; see Definition 2.1.4. For $x \in \Omega$, the **orbit** $\mathcal{O}_x = Gx$ of G through x is the subset of Ω defined by

$$\mathcal{O}_x = \varphi^{-1}(\varphi(x)).$$

Thus the G -orbits are the equivalence classes for the relation $\sim_G$; see Remark 2.1.5.

Equivalently, we have $\mathcal{O}_x = \{gx \mid g \in G.\}$

Proposition 2.3.6: Ω is the disjoint union of the G -orbits in Ω .

Proof: This follows from Proposition 2.1.3. ■

Remark 2.3.7: Each orbit $\mathcal{O}_x$ is itself a G -set.

2.4. Quotients of groups

Let G be a group and let H be a subgroup of G . There is an action of H on the set G by right multiplication: for $h \in H$ and $g \in G$ we can define $h \cdot g = gh^{-1}$.

We are going to consider the quotient of G by the equivalence relation of H -conjugacy; this equivalence relation is defined by

$$g \sim g' \Leftrightarrow \exists h \in H, g = g'h.$$

Definition 2.4.1: The **left quotient of G by H** is the quotient $(\pi, G/H)$ of G by the equivalence relation of H -conjugacy defined using the action of H on G by right multiplication as described above.

Remark 2.4.2:

- Of course, one can use an explicit model for the quotient by taking G/H to be the set of equivalence classes in G for the H -conjugacy relation.
- The equivalence classes for the relation of H -conjugacy defined by the action of right multiplication are precisely the **left cosets** of H in G . The class of $x \in G$ has the form

$$xH = \{xh \mid h \in H\}$$

.

For $x \in G$,

$$\pi^{-1}(\pi(x)) = xH.$$

- We can also consider the action of H on G by left multiplication. This action determines an equivalence relation of H -conjugacy, and the quotient of G by this equivalence relation is called the **right quotient of G by H** and is written $(\pi, H \backslash G)$. In this case, the equivalence classes are the **right cosets** where the class of $x \in G$ has the form $Hx = \{hx \mid h \in H\}$.

For $x \in G$, we have $\pi^{-1}(\pi(x)) = Hx$.

Proposition 2.4.3: There is an action

$$\alpha : G \times G/H \rightarrow G/H$$

of the group G on the set G/H such that

$$\forall g, x \in G, \text{ we have } \alpha(g, \pi(x)) = \pi(gx)$$

where $\pi : G \rightarrow G/H$ is the quotient map.

Proof: To define the action map α , first fix $g \in G$. We are going to define the mapping

$$\alpha(g, -) : G/H \rightarrow G/H.$$

Consider the mapping $\pi_g : G \rightarrow G/H$ given by the rule $\pi_g(x) = \pi(gx)$. This mapping has the property that $x \sim_H x' \Rightarrow \pi_g(x) = \pi_g(x')$. Indeed,

$$x \sim_H x' \Rightarrow \exists h, x = x'h \Rightarrow \pi_g(x) = \pi(gx) = \pi(gx'h) = \pi(gx') = \pi_g(x')$$

by the defining property of π ; see Definition 2.1.4. Again using Definition 2.1.4 we find the desired mapping $\alpha(g, -) : G/H \rightarrow G/H$ with the property that

$$(\clubsuit) \quad \alpha(g, -) \circ \pi = \pi_g.$$

We now assemble the mappings $\alpha(g, -)$ to get a mapping $\alpha : G \times G/H \rightarrow G/H$ which satisfies $\alpha(g, \pi(x)) = \pi(gx)$ for each $g, x \in G$, and it remains to check that α determines an action as in Definition 2.3.1.

Of course, using $(\clubsuit)$, we have $\alpha(1, -) \circ \pi = \pi_1 = \pi = \text{id} \circ \pi$; since π is surjective, it follows that $\alpha(1, -) = \text{id}$. Thus $\alpha(1, z) = z$ for each $z \in G/H$, which shows that α satisfies the first requirement of Definition 2.3.1.

Now suppose that $g_1, g_2 \in G$. To complete the proof, we must verify the remaining requirement of Definition 2.3.1; thus we must show that

$$(\heartsuit) \quad \alpha(g_1, \alpha(g_2, -)) = \alpha(g_1 g_2, -)$$

On the one hand, using $(\clubsuit)$ we find that

$$\alpha(g_1 g_2, -) \circ \pi = \pi_{g_1 g_2};$$

on the other hand, for $z \in G$ we have

$$\begin{aligned} (\alpha(g_1, \alpha(g_2, -)) \circ \pi)(z) &= \alpha(g_1, \alpha(g_2, \pi(z))) \\ &= \alpha(g_1, \pi_{g_2}(z)) && \text{by } (\clubsuit) \\ &= \alpha(g_1, \pi(g_2 z)) \\ &= \pi(g_1(g_2 z)) && \text{by } (\clubsuit) \end{aligned}$$

Since π is surjective, $(\heartsuit)$ follows at once. This completes the proof. ■

2.5. Quotients of groups and orbits.

Definition 2.5.1: Suppose that G acts on Ω_1 and on Ω_2 . A morphism of G -sets $\varphi : \Omega_1 \rightarrow \Omega_2$ is a function φ with the property that $\forall g \in G$ and $\forall x \in \Omega_1$, we have $\varphi(gx) = g\varphi(x)$.

The morphism of G -sets φ is an isomorphism (of G -sets) if there is a morphism of G -sets $\psi : \Omega_2 \rightarrow \Omega_1$ such that $\varphi \circ \psi = \text{id}$ and $\psi \circ \varphi = \text{id}$.

Suppose that G acts on Ω and let $x \in \Omega$.

Definition 2.5.2: The stabilizer of x in G is the subgroup $\text{Stab}_G(x) = \{g \in G \mid gx = x\}$.

Proposition 2.5.3: Write $H = \text{Stab}_G(x)$ and recall that $\pi : G \rightarrow G/H$ is the quotient mapping. There is a unique isomorphism of G -sets $\gamma : G/H \rightarrow \mathcal{O}_x$ with the property that

$$\gamma(\pi(1)) = x.$$

Proof: The rule $g \mapsto gx$ determines a surjective mapping $\alpha_x : G \rightarrow \mathcal{O}_x$. Recall that the action of H on G by right multiplication determines an equivalence relation $\sim$ on G used to construct the quotient G/H .

For $g_1, g_2 \in G$ we find that

$$g_1 \sim g_2 \Rightarrow \exists h \in H, g_1 = g_2 h \Rightarrow \alpha(g_1) = \alpha(g_2 h) = g_2 h x = g_2 x = \alpha(g_2)$$

since $h \in H = \text{Stab}_G(x) \Rightarrow hx = x$.

Thus Definition 2.1.4 shows that there is a mapping $\gamma : G/H \rightarrow \mathcal{O}_x$ such that $\gamma \circ \pi = \alpha_x$. To see that γ is a morphism of G -sets, it suffices to show that $(\clubsuit) \quad \forall g, g' \text{ we have}$

$$\gamma(g \cdot \pi(g')) = g \cdot \gamma(\pi(g')).$$

Now by the definition of the G -action on G/H we have $g \cdot \pi(g') = \pi(gg')$; see Proposition 2.4.3. Thus $\gamma(g \cdot \pi(g')) = \gamma(\pi(gg')) = \alpha_x(gg') = gg' \cdot x$. On the other hand, $g \cdot \gamma(\pi(g')) = g \cdot \alpha_{x(g')} = g \cdot g' \cdot x$ which confirms $(\clubsuit)$. This shows that γ is indeed a morphism of G -sets.

Since α_x is surjective and $\gamma \circ \pi = \alpha_x$, also γ is surjective. It only remains to see that γ is injective. Suppose that $z, z' \in G/H$ such that $\gamma(z) = \gamma(z')$. Since $\pi : G \rightarrow G/H$ is surjective, we may choose $g, g' \in G$ with $z = \pi(g)$ and $z' = \pi(g')$. Now

$$\gamma(z) = \gamma(z') \Rightarrow \gamma(\pi(g)) = \gamma(\pi(g')) \Rightarrow \alpha_{x(g)} = \alpha_{x(g')} \Rightarrow gx = g'x.$$

We now conclude that $g^{-1}gx = x$ so that $g^{-1}g \in \text{Stab}_G(x) = H$. Since the quotient mapping π is constant on H -orbits, $z = \pi(g) = \pi(gg^{-1}g') = \pi(g') = z'$. This shows that γ is injective and completes the proof. ■

Definition 2.5.4: The action of G on Ω is **transitive** if there is a single G -orbit on Ω . Equivalently, the action is transitive if the quotient $\Omega / \sim$ is a singleton set.

Example 2.5.5: Let I be a set and let $G = S(I)$ be the group of permutations of I . Fix $x \in I$ and let $H = \text{Stab}_G(x)$. Notice that G acts on I . Moreover, the G -orbit of x is precisely I - in other words, the action of G on I is transitive.

Notice that $H = S(I - \{x\})$.

Now Proposition 2.5.3 gives an isomorphism of G -sets $G/H \rightarrow I$; i.e. $S(I)/S(I - \{x\}) \rightarrow I$.

2.6. Products of subgroups and symmetric groups

Definition 2.6.1: If $H, K \subseteq G$ are subgroups, H **normalizes** K if for each $g \in H$ we have $gKg^{-1} \subseteq K$, where $gKg^{-1} = \{gxg^{-1} \mid x \in K\}$.

Definition 2.6.2: Let H, K be subsets of G . The **product** of H and K is

$$HK := \{xy \mid x \in H, y \in K\}.$$

For homework, you will prove

Proposition 2.6.3: Suppose that H, K are subgroups of G and that H normalizes K . Then $\langle H, K \rangle = HK$. In particular, HK is a subgroup of G .

Remark 2.6.4: It's often useful to view the symmetric group S_n as $S(\mathbb{Z}/n\mathbb{Z})$, the permutations of the integers mod n . Once you do, you get to **compute** with indices using ordinary arithmetic in $\mathbb{Z}/n\mathbb{Z}$ – addition, negation, etc. – rather than tracking permutations by hand. This can be a useful point-of-view.

Example 2.6.5: View $S = S_n$ as $S(\mathbb{Z}/n\mathbb{Z})$. Define $\sigma, \tau \in S$ by

$$\sigma(i) = i + 1 \text{ and } \tau(i) = -i,$$

with the arithmetic taking place in $\mathbb{Z}/n\mathbb{Z}$. As permutations, σ is the n -cycle $(1, 2, \dots, n)$ and τ is the product of transpositions $\prod_{i=1}^{\lfloor \frac{n}{2} \rfloor} (i, n - i)$; in particular σ has order n and τ has order 2.

We check the normalizer condition $\tau\langle\sigma\rangle\tau^{-1} \subseteq \langle\sigma\rangle$ directly: for $i \in \mathbb{Z}/n\mathbb{Z}$,

$$(\tau\sigma\tau)(i) = \tau(\sigma(\tau(i))) = \tau(\sigma(-i)) = \tau(-i + 1) = i - 1 = \sigma^{-1}(i),$$

so $\tau\sigma\tau = \sigma^{-1} \in \langle\sigma\rangle$. Thus $\langle\tau\rangle$ normalizes $\langle\sigma\rangle$, and Proposition 2.6.3 shows

$$\langle\sigma, \tau\rangle = \langle\sigma\rangle\langle\tau\rangle$$

is a subgroup of S . We call $D = \langle\sigma, \tau\rangle$ the **dihedral group** (of order to be determined – see the problem set).

2.7. Lagrange's Theorem

Let H be a subgroup of the group G and write G/H for the (left) quotient, as above. Recall that the H -cosets xH are the H -orbits for this action.

Theorem 2.7.1: There is a bijection $\varphi : (G/H) \times H \rightarrow G$ for which $\{\varphi(z, h) \mid h \in H\}$ is an H -orbit (i.e. a left H -coset) for each $z \in G/H$.

Proof: Indeed, using the axiom of choice we select for each $z \in G/H$ an element $g_z \in \pi^{-1}(z)$ where $\pi : G \rightarrow G/H$ is the quotient map.

Now define $\varphi : (G/H) \times H \rightarrow G$ by the rule $\varphi(z, h) = g_z h$.

To see that φ is onto, let $g \in G$. One then knows that $g \sim g_z$ for some $z \in G/H$. Since $\pi^{-1}(z) = g_z H$ it follows that $g = g_z h$ for some $h \in H$, so $g = \varphi(z, h)$.

To see that φ is injective, suppose that $\varphi(z, h) = \varphi(z', h')$. Then $g_z h = g_{z'} h'$ so that

$$(g_{z'})^{-1} g_z \in H \Rightarrow g_z \sim g_{z'} \Rightarrow z = z'.$$

Now $g_z h = g_{z'} h' \Rightarrow h = h'$ which completes the proof that φ is injective. The remaining assertion follows from the definition of φ . ■

Corollary 2.7.2: Suppose that G is a finite group and that H is a subgroup of G . Then

$$|G| = |G/H| \cdot |H|.$$

Proof: Indeed, for finite sets X and Y , we have $|X \times Y| = |X| |Y|$. ■

